

Fungal infection 2

Dr eman gomaa

Consultant dermatologist

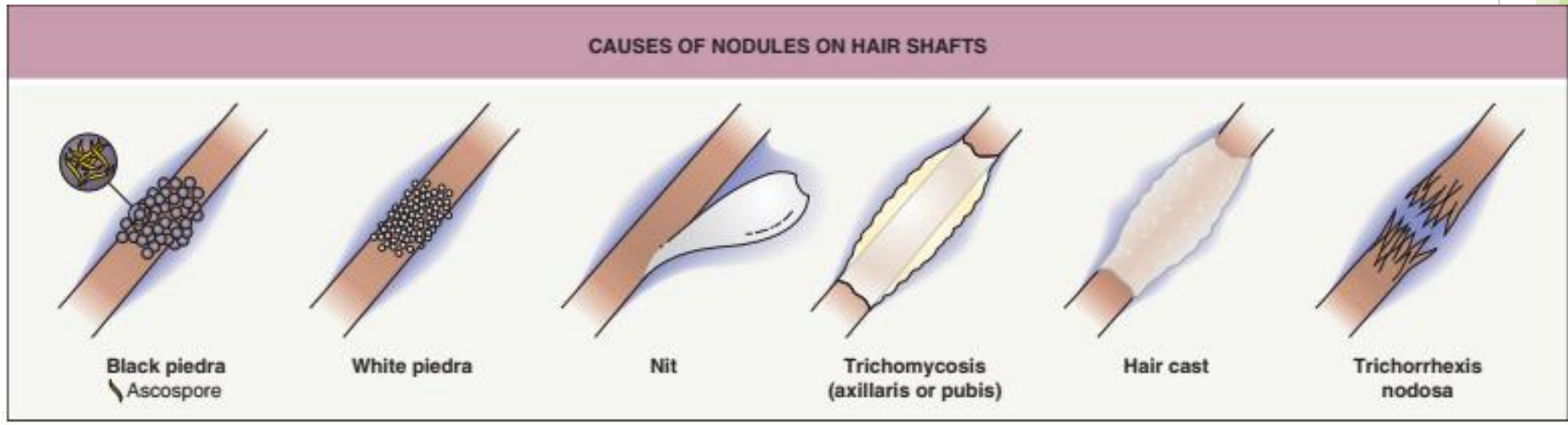
Other moulds

Piedra

	Black piedra	White piedra
	<i>Piedraua hortae</i>	<i>Trichosporon beigeli</i>
	Soil , no human to human	
	Nodule →black , hard , firmly adherent	Nodule :white ,soft , loosa
	scalp	, face ,axilla, pubic region ,may be scalp
KOH	Brown septated hyphae	Non demataceous hyphae
TT	Shaving the hair, shampoo , topical antifungal	



- ▶ DD: nodules of hair shaft
 - ▶ Nits , haircast , trichomycosis axillaris , trichorrhexis nodosa .



COMPARISON OF BLACK AND WHITE PIEDRA

	White piedra	Black piedra
Nodule color	White (occasionally red, green, or light brown)	Brown to black
Nodule firmness	Soft	Hard
Nodule adherence to the hair shaft	Loose	Firm
Typical anatomic location	Face, axillae, and pubic region (occasionally scalp)	Scalp and face (occasionally pubic region)
Favored climate	Tropical	Tropical
Causative organism	<i>Trichosporon</i> spp. [*] , e.g. <i>T. ovoides</i> , <i>T. inkin</i>	<i>Piedraia hortae</i>
KOH examination of "crush prep" of cut hair shafts	Non-dematiaceous hyphae with blastoconidia and arthroconidia	Dematiaceous hyphae with asci and ascospores [†]
Culture on Sabouraud's agar	Moist, cream-colored, yeast-like colonies [‡]	Slow-growing, dark green to dark brown-black colonies
Treatment	Clip affected hairs, wash with antifungal shampoo	Clip affected hairs, wash with antifungal shampoo

Malassezia

Dermatosis caused by Malassezia

TVC

SD

PS

SEBO -PS

Atopic dermatitis

Pityrosporum folliculitis

Neonatal acne

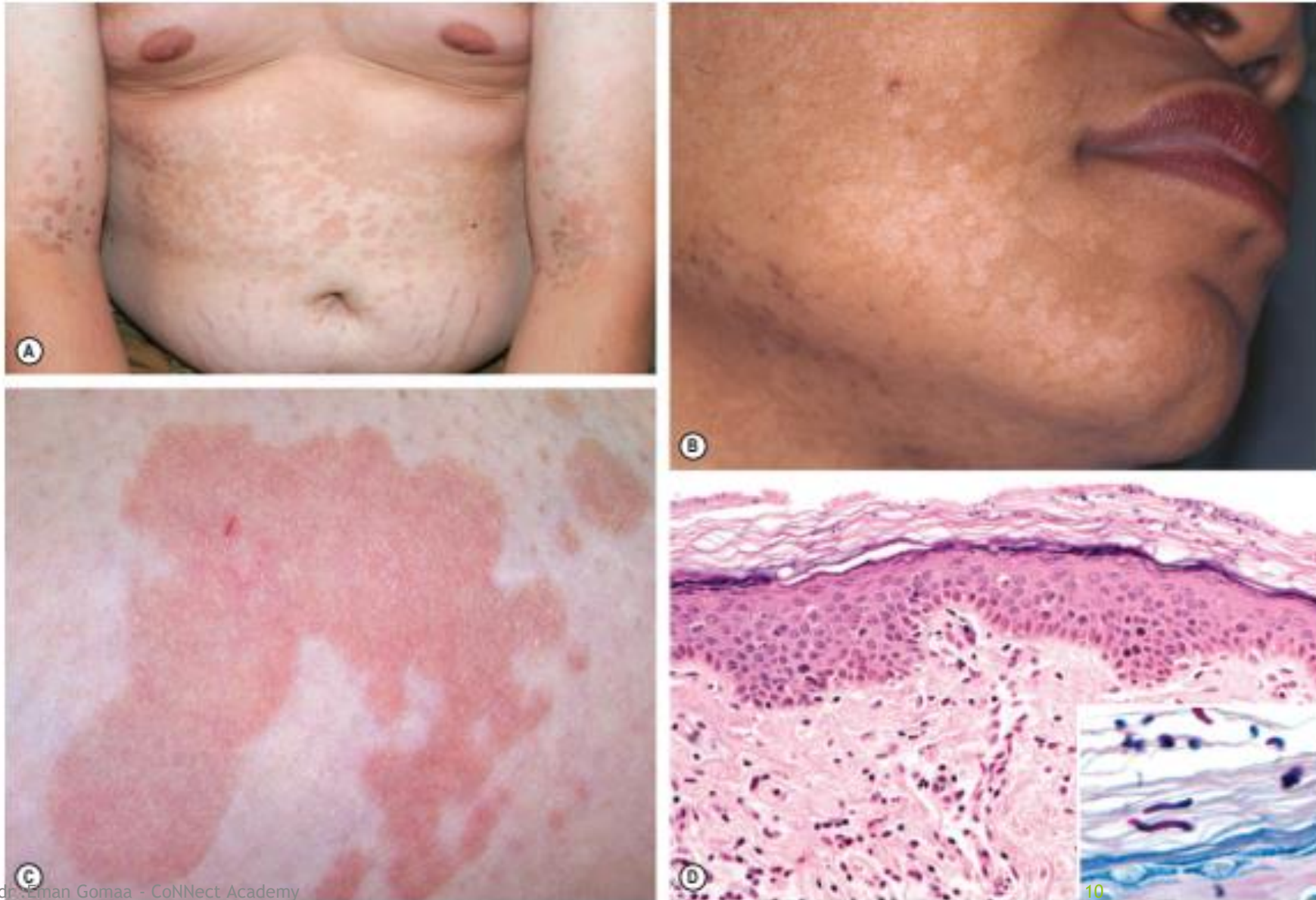
Confluent , reticulat papilomatosis

Onychomycosis

Ptyriasis versicolor

- ▶ Mild chronic superficial fungal infection
- ▶ M.GLOBOSA
- ▶ M .FURFUR
- ▶ M. SYMPODIALIS
- ▶ Commensal →yeast
- ▶ Pathogenic →moulds

- ▶ Adult → due to increase sebum
- ▶ Upper trunk
- ▶ Asymptomatic welldefined oval patch
- ▶ Pruritis → lipase
- ▶ Types :
 - ▶ Hypopigmented
 - ▶ Hyperpigmented
 - ▶ Mixed
 - ▶ Erythematous
 - ▶ perifollicular



SCE Dermatology / EBDV by dr. Eiman Gomaa - CoNnect Academy

Fig. 77.3 Tinea (pityriasis) versicolor. A Hyperpigmented variant with confluence on the lower chest. B Hypopigmented variant on the face. C Coalescing pink lesions with fine scale. D Yeast and short hyphal forms in the stratum corneum highlighted by a PAS stain (inset). A, B, Courtesy, Kalman Watsky, MD; C, Courtesy, Lorenzo Cerroni, MD.

► DD:

► Hypopigmented patch on trunk :

► Vitiligo , p .alpa , leprosy

► Hyperpigmented :

► Pih , mf , ash dermatosis

► Erythematous :

► SD ,PR , 2RY SYPHILIS , t. cruris

Investigation

- ▶ Woods light :
 - ▶ Golden yellow fluorescence
- ▶ Microscopic :
 - ▶ Spaghetti , meat ball
- ▶ Cultur :
 - ▶ SDA WITH OLIVE OIL
 - ▶ DIXON AGAR
 - ▶ LN (leeming not man)
 - ▶

► Pathogenicity :

- Lipase →FFA→irritation → pruritis
- Azelaic acid →antimicrobial
- Decrease tyrosinase →hypopigmentation

TREATMENT OF TINEA (PITYRIASIS) VERSICOLOR

Initial therapy (often combination)	Topical* • Antifungal shampoo applied as a body cleanser for 10–15 minutes before rinsing, twice weekly for 2–4 weeks - Selenium sulfide shampoo, 1% (OTC) or 2.5% - Ketoconazole shampoo, 1% (OTC) or 2% - Zinc pyrithione shampoo - Ciclopirox olamine shampoo, 1% • Antifungal cream applied several hand-widths beyond clinically visible lesions, daily to twice daily for 2 weeks - Imidazole, e.g. ketoconazole 2% cream - Ciclopirox olamine 0.77% cream, gel, or lotion Oral** • Fluconazole: 200 mg po daily x 5–7 days[†]; 200–300 mg weekly x 2–3 weeks; or 400 mg once • Itraconazole: 200 mg po daily x 5–7 days
Maintenance therapy to prevent recurrence	• Treat previously affected sites with topical imidazole daily for 2 weeks prior to anticipated sun exposure • Apply antifungal shampoo (see above) 1–2 times weekly

onychomycosis

- ▶ Fungal infection of nail plat , bed
- ▶ aetiopathogenesis

dermatophyte	Non dermatophytes	yeast
T rubrum T.Mentagrophyte T. floccosum	Scytalidium Fusarium	Candidiasis Finger nail



Main sign

1. Subungual hyperkeratosis
2. Nail discoloration
3. Onycholysis
4. Paronychia



Clinical variants

Distal ,lateral subangual , onychomycosis	Porximal subangual	White superficial	Black superficial
Commest T rubrum Enter nail through hyponychium T . Pedis	Rare t. Rubrum Sign of HIV Non dermatophyt Aspergillus Fusirium	White opaque Easly scraped T mentagrophyte	T.Rubrum var -nigricans scytalidium dimidiatum Patchy or transverse



endonyx	Total dystrophic	Candidal
All thickness of nail T. Soudnans nail plate only No nail ped No hyperkeratosis No onycholysis Milky white	Most sever Long standing DLSO PSO	CMC PARONYCHEA



Fungal toenail classifications



Distal lateral
subungual
onychomycosis



Superficial white
onychomycosis



Proximal subungual
onychomycosis



Endonyx
onychomycosis



Total dystrophic
onychomycosis

INDICATORS OF MORE SEVERE ONYCHOMYCOSIS WITH A POOR RESPONSE TO TREATMENT

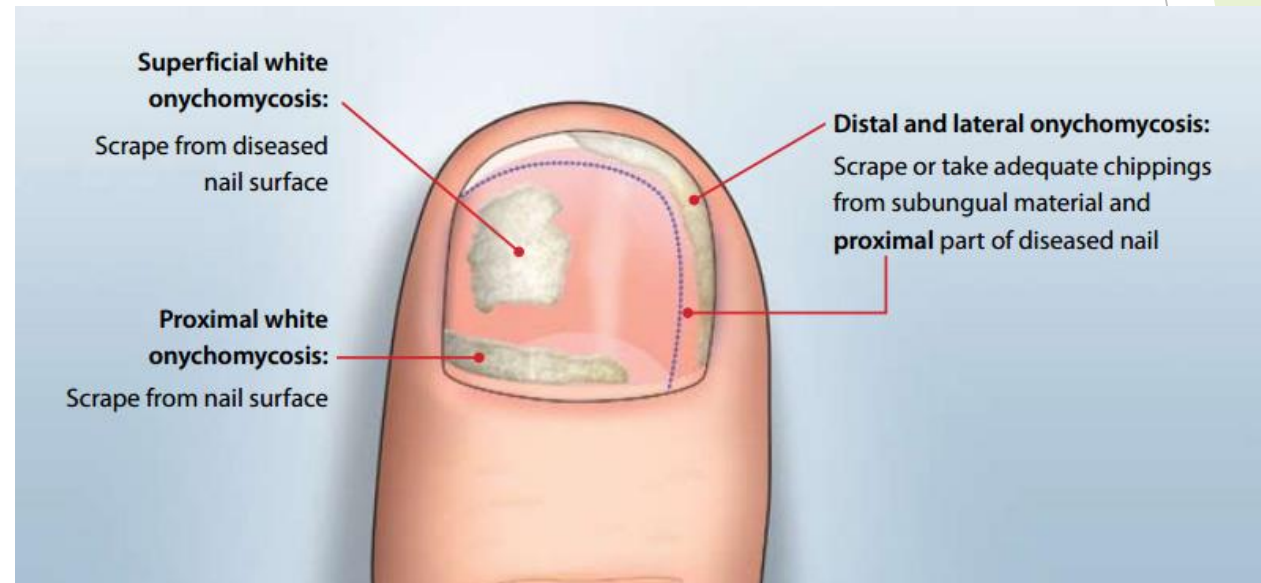
Nail factors

- Subungual hyperkeratosis >2 mm thick[★]
- Significant lateral involvement
- Dermatophytoma[†]
- >50% involvement of nail bed
- Slow nail growth rate
- Total dystrophic onychomycosis
- Matrix involvement

Patient factors

- Immunosuppression
- Peripheral arterial disease
- Poorly controlled diabetes mellitus

- ▶ Sampling :
 - ▶ DLSO :subungual debris
 - ▶ Pso :punch biopsy
 - ▶ Wso , BSO : scraping
 - ▶ EO → clipping
- ▶ Direct microscopy :
 - ▶ KOH 20%
- ▶ CULTURE



DD

DLSO

Traumatic
Nail ps

Pso

Bacterial paronychia
Pustular ps of nail

WSO→

TRAUMATIC
LEUKONYCHIA

TREATMENT

SYSTEMIC
Terbinafin
Itraconazol
Fluconazol

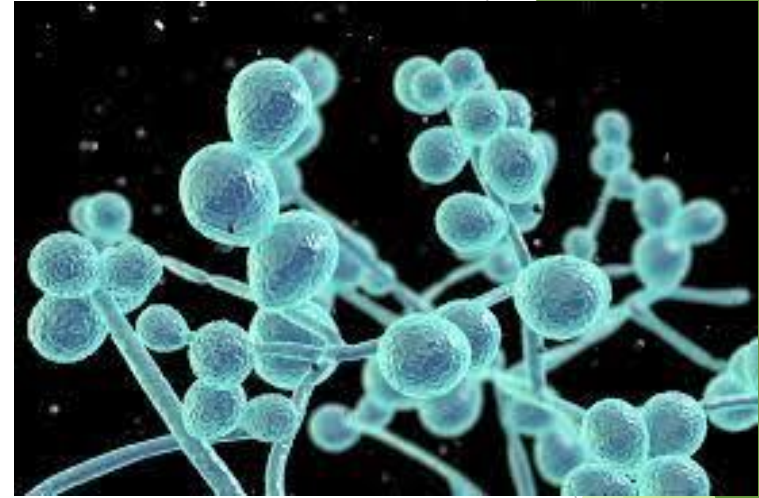
TOPICAL
Ticonazol 28%

LASER
Co2 laser
Laser assisted
therapy
Long pulsed
ND- YAG

candidiasis

► Taxonomy

- Genus candida include over 100 species



Candida albicans	Non - albicans
Most common Polymorphisim Budding , pseud hyphae	1. C-tropicalis 2. C- pseudotropicalis 3. C-krusei 4. C-glaborata

Is candida normal flora ??

► Candida is normal commensal flora

1. Gut : c- albicans increase by AB , hospitalization
2. Vaginal :c-albicans , c- glabrata ,
 1. Increase by pregnancy , ocp , iud , hospitalization
3. Skin :normaly not present
 1. Skin adjacent to orifices , skin of fingers
 2. Increase by age , climate
 3. May persist in moist intertriginous

Aetiopathogenesis

Source:- commensal yeast

❑ Predisposing factors:-

- a) A.B., pregnancy, **O**CP
- b) Immune suppressi**o**n, DM, HIV,
- c) Xer**o**stomia, **o**ral denture.
- d) O**cclusion hyperhidrosis.

❑ Pathogenesis:-

❑ Fungal virulence factor:-

Species:- C.Albicans. Adherence. Toxic enzymes. Melanin production

❑ Host factors:-

Local tissue damage.

Competition between candida – bacterial flora.

Clinical feature

a) Oral.(8)

b) Genital.(2)

c) Skin.(4)

D) NAIL 2

E) others

□ Oral:-

► 1) **oral thrush**:-

► Acute pseudomembranous candidiasis.

1st week of life in preterm.

HIV & steroid.

C.Pic:- well defined patches of creamy crumbly white pseudomembrane.

With erythematous base → gum, palate & buccal.

H.p.:- desquamated epithelium, mycelium, fibrin & leukocyte.

Complications:- esophagus, erosion, ulcer & HS



2) Candidal leukoplakia

- ▶ Chronic plaque like chronic hyperplasia P.F.
 - ▶ Chronic muco-cutaneous candida → smoking.
 - ▶ C.pic:- male > 30 ys with
 - mild soreness & roughing.
 - persistent firm irregular white plaque → tongue.
- DD:- oral thrush & other leukoplakia.
- Comp:- malignancy.



3)Angular cheilitis

- ▶ Perleche.
- ▶ Soreness, cracking at angle of mouth.
- ▶ Intertigo most commonly caused by candida.
- ▶ Nutritional status.
- ▶ mechanical.
- ▶ Moisture → dentures stomatitis



4) Acute erythematous

- ▶ AB.
 - ▶ Acute erythematous atrophic dorsum of the tongue.
-

5)Chronic pseudomembranous.

6)Chronic erythematous.

7)Chronic nodular.

8)Median rhomboid glossitis.



Gentital vuvlo-vaginal

- ▶ R.F:- pregnancy, premenstrual, OCP, IUD & sexually active.
- ▶ 75% of women.
- ▶ C.pic:- itching, soreness, thick creamy white scanty vaginal discharge.
- ▶ DD:- Vaginal discharge PS, contact & LSEA
- ▶ Physiological cervical ectopy → trichomonas & chlamydia.



Figure 2. Genitalia with candidiasis, showing swollen labias and reddened vagina.

Candidal palanitis:- skin of glans penis → uncircumsized sexual partner.

C.Pic:- tiny papule & pustules → glans penis → soreness & irritation.

DD:- bacterial HS, PS, LP, LSEA, plasma cell palanitis & Reiter.



B -skin

- ▶ 1) candida intertrigo :any skin fold
- ▶ 2)Napkin candidiasis:- C.Albicans - oral tropical AB & steroids.
 - DD:-

Common :- Contact dermatitis & SD

Less common:- bullous impetigo, strept infection, strept perianal dermatitis, SPS & atopic dermatitis.

Rare:- acrodermatitis enteropathica , letterer siwe disease.

3)Granuloma gluteal infantum

- ▶ Nodular granulomatous candidiasis.
- ▶ Violaceous nodule & plaque at buttocks and genitalia.
- ▶ Reminiscent of Kaposi.
- ▶ Reactive condition to severe contact dermatitis.
- ▶ Diarrhea & urine incontinence.
- ▶ Jacquet erosive → pushed out ulcer.



4) Perianal scrotal candidiasis

- ▶ Anal margin involve scrotum

Nail (2)

- ▶ 1) Paronychium :
- ▶ affect nail fold.
- ▶ Poor peripheral circulation.
- ▶ P.F.:- hydration, water, flour, CHO.
- ▶ C.Pic:- affects several fingers → Toe nails aren't affected
- ▶ Red swollen proximal nail folds → Pain & tenderness.
- ▶ Loss of cuticle → pocketing.
- ▶ Complications :- dystrophy, discoloration and onycholysis

► 2) Onychomycosis:

- Affects nail plate.
- DLSO.
- Complete destructive erosion



Fig. 77.20 Mucocutaneous candidiasis. A Thrush with "cottage cheese"-like exudate on the buccal mucosa. B Thrush and candidal cheilitis. C Angular cheilitis (perleche). D Candidiasis of the suprapubic area and penis in a young boy. Note the collarettes of scale on the coalescing, brightly erythematous papules. E Candidiasis of the scrotum and medial thighs with beery red erythema, scale, and satellite papules. F This infection occurred in a hospitalized patient with diabetes mellitus who was receiving broad-spectrum antibiotics. Note the multiple satellite lesions. G Erosive interdigital candidiasis (erosio interdigitalis blasmomycetica) in the classic location between the third and fourth fingers. H Dermal candidiasis in an immunosuppressed patient. A, Courtesy, John Lewis, MD; B, C, Courtesy, David A. Rapaport, MD; D, E, Courtesy, David A. Rapaport, MD; F, G, Courtesy, David A. Rapaport, MD; H, Courtesy, David A. Rapaport, MD.

Others Congenital Chronic congenital candidiasis

- ▶ Congenital:- candida of the skin (birth membrane) at time of birth.
- ▶ PF:- prematurity & IUD.
- ▶ C.pic:- turbid fluid ,discrete vesicles, pustules on erythematous base in face & chest

Investigations

- Microscopic:- skin, nail, oval yeast - bud & pseudo and true hyphae.
 - Culture:-
- ✓ Cyclohexamid free 37 degree centigrade
- ✓ Rapid grower 1-3 days.
- ✓ Glucose pepton agar :- white, creamy & soft.
- ✓ Chromogenic agar.
- Germ tube test→ rapid screening test.
- Histopathology:- PAS-GMS.

TTT

❑ General :-

- removal of predisposing factor.
- mouth:- good hygiene & Skin:- careful drying.

❑ Therapeutic:

a) Topical :- nystatin, imizdazol & clotrimazol.

b) Systemic:

-fluconazole & itraconazol (most useful).

-New:- voriconazol, posaconazol.

R/Caspofungin IV fungal cell wall inhibitor.

❑ TTT of special type.

TREATMENT OF MUCOCUTANEOUS CANDIDA INFECTIONS	
Type of infection	Treatment regimen (adult doses unless otherwise indicated)
Oropharyngeal candidiasis	Nystatin* 100 000 units/ml suspension: • Children and adults – 4–6 ml swish and swallow four times daily • Infants – 2 ml (1 ml inside each cheek) four times daily Clotrimazole* 10 mg troche five times daily Fluconazole† 200 mg po on day 1, then 100–200 mg po daily Continue treatment for 7–14 days after clinical resolution
Esophageal candidiasis	Fluconazole* 200–400 mg po on day 1, then 100–400 mg daily Itraconazole 200 mg po daily Voriconazole 200 mg po or iv BID Posaconazole 400 mg po BID Caspofungin 50 mg po iv daily Continue treatment for 7–14 days following resolution of symptoms, for a minimum of 21 days total
Candidal vulvovaginitis ⁵	Fluconazole* 150 mg po single dose or (for severe disease or an immunosuppressed patient) three doses at 3-day intervals Butoconazole 2% vaginal cream, 5 g daily for 1–3 days Clotrimazole*,:‡ • 1% vaginal cream (or other topical formulation), 5 g daily for 7–14 days • Vaginal suppositories: 100 mg daily for 7 days or 200 mg daily for 3 days Miconazole*,:‡ • 2% cream, 5 g daily for 7 days • Vaginal suppositories: 100 mg daily for 7 days, 200 mg daily for 3 days, or 1200 mg single dose Tioconazole* 6.5% cream, 5 g single dose Terconazole* • 0.4% or 0.8% cream, 5 g daily for 7 or 3 days, respectively • 80 mg vaginal suppository daily for 3 days Nystatin 100 000 units vaginal suppository daily for 14 days <i>For suppression in patients with recurrent infections:</i> Fluconazole 150 mg weekly for 6 months Clotrimazole vaginal suppository, 200 mg twice weekly or 500 mg weekly for 6 months
Chronic mucocutaneous candidiasis	Fluconazole • <i>Eradication:</i> 400–800 mg po daily for 4–6 months • <i>Suppressive therapy:</i> 200 mg po daily
Candidal intertrigo or balanitis ⁵	Topical imidazole (see Table 127.12) or ciclopirox cream or lotion twice daily for 1–2 weeks or until resolved[†] Systemic agents for recalcitrant or severe cases: Fluconazole • 50 to 100 mg po daily for 14 days • 150 mg po weekly for 2–4 weeks Itraconazole 200 mg po twice daily for 14 days

Subcutaneous fungal infections

Dermis, S.C, muscle, fascia

❑ Mycetoma

Madura foot

Def:- localized chronic granulomatous S.C. fungal infection

Subtypes:

a)Fungal (eumycotic):- *Madurella mycetosis* & *Madurella grisea*

b)Actinomycotic:- *Nocardia* → *brasilliance*, *asteroids* & *caviae*.

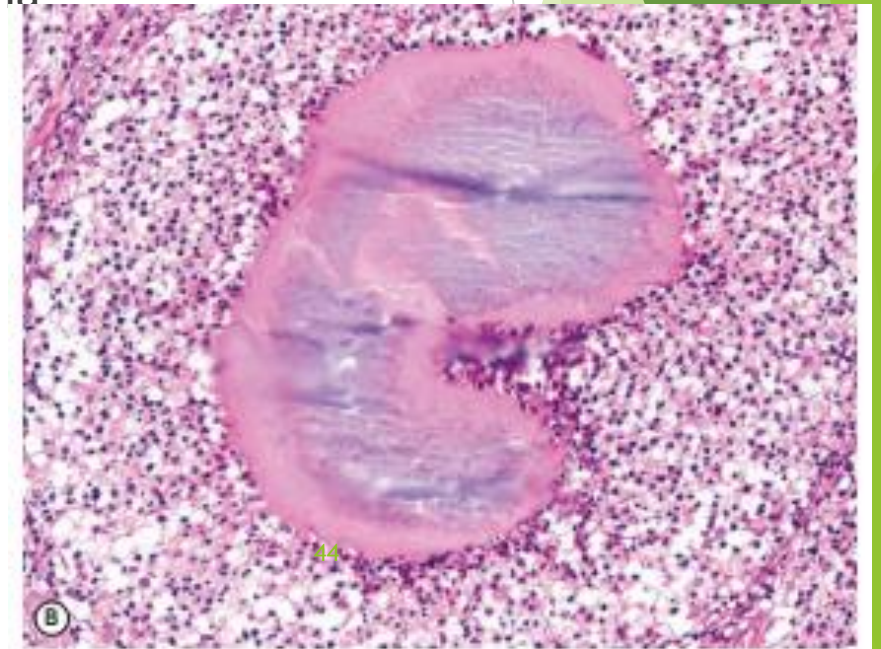
Actinomyces:- *Israeli*, *Madura* & *pelletieri* (*actinomadura*)

- ❑ Source:- Soil, agriculture worker without foot wear → direct inoculation to skin.
- ✓ Foot unilateral
- ✓ Painless papule→local edema.
- ✓ Purulent sinuses→ sulfer granule sclerotia → compact masses of fungal colonies.
- ✓ Very slowly 20-30 yrs after the initial injury.



□ Investigations:-

- ▶ Radiological studies:- X-ray, C.T. & MRI → degree of involvement.
 - ▶ Microscopic pic of pus:- grains like neutrophils.
 - ▶ H.P.:- pseudoepitheliomatous hyperplasia, suppurative granuloma
- *Granules→ tightly packed colonies. (PAS & methanamine silver).



COLORS OF GRAINS AND GEOGRAPHIC DISTRIBUTION OF EUMYCOTIC AND ACTINOMYCOTIC MYCETOMAS

Pathogen		Grain color	Geographic distribution
Eumycotic	<i>Madurella mycetomatis</i>	Black	North, Central and South America; Caribbean, Africa, Europe, Middle East, Asia
	<i>Madurella grisea</i>	Black	North, Central and South America; Africa, Asia
	<i>Leptosphaeria senegalensis</i>	Black	Africa, Asia
	<i>Pseudallescheria boydii</i> /Scedosporium apiospermum* (sexual/asexual states, respectively)	White	North, Central and South America; Africa, Oceania, Asia, Europe
	<i>Acremonium</i> and <i>Sarocladium</i> spp.	White	North, Central and South America; Asia, Europe, Oceania
Actinomycotic	<i>Nocardia brasiliensis</i> †	White	Worldwide (all <i>Nocardia</i> spp.)
	<i>Nocardia asteroides</i> ‡	White	
	<i>Nocardia caviae</i>	Yellow-white	
	<i>Actinomyces israeli</i> §	Yellow-white	Worldwide
	<i>Actinomadura madurae</i>	Pink or white	Worldwide
	<i>Actinomadura pelletieri</i>	Red	North and South America, Africa, India
	<i>Streptomyces somaliensis</i>	Brown or yellow	Africa (arid regions)

TTT

► Combined surgical & medical.

Eumycotic:- itraconazol

Actinomyotic:- Amikacin + trimethoprim.

Sporotrichosis

Rose gradner

- ▶ *Sporothrix schenckii*.
- ▶ Soil, gradner & brazil.
- ▶ Hand → drained by lymph (sporotrichosis)
- ▶ Single painless papule → ulceration.
- ▶ Lymphatic spread → up to arm.
- ▶ Subsequent exposure → no lymphatic spread.
- ▶ Cigar shaped organism → budding yeast.
- ▶ Suppurative granuloma → PAS & Silver.
- ▶ ttt → itraconazol.
- ▶ DD → VI

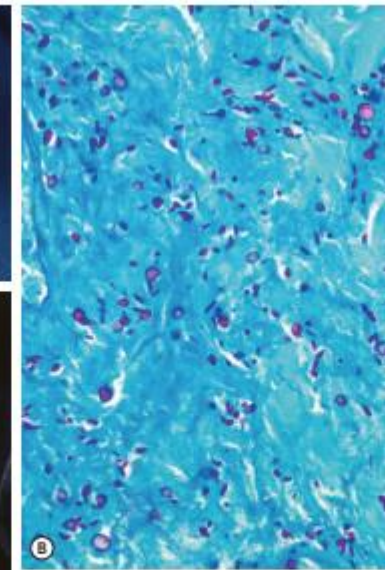


Fig. 77.24 Lymphocutaneous (sporotrichoid) pattern. A An eroded nodule on the thumb representing the primary lesion with a secondary lesion along the lymphatics due to sporotrichosis. B Multiple yeast forms of *Sporothrix schenckii* in the dermis; note that many of the organisms have the characteristic cigar shape. C Ulcerated nodule on the extensor forearm with multiple more proximal nodules due to nocardiosis in a patient with lymphoma who was receiving systemic corticosteroids. B, Courtesy, Ronald P. Rapini, MD; C, Courtesy, Jean L. Bolognia, MD.

INFECTIOUS CAUSES OF A LYMPHOCUTANEOUS ("SPOROTRICHOID") PATTERN

Most common

Atypical mycobacteria, especially *M. marinum*, but also other species (e.g. *M. chelonae*, *M. kansasii*)

Sporotrichosis (Fig. 77.24A)

Unusual

Nocardiosis (Fig. 77.24C)

Pyogenic bacteria (e.g. *Staphylococcus aureus*, *Streptococcus pyogenes*)

Pseudallescheria boydii/*Scedosporium apiospermum*

Rare (in high-income countries)

Leishmaniasis

Tularemia*

Tuberculosis*

Dimorphic fungi (other than *Sporothrix* spp.)

Opportunistic fungi in immunocompromised hosts (e.g. *Fusarium*, *Alternaria* spp.)

Glanders (*Burkholderia mallei*)*

Cat scratch disease*

Anthrax

Cowpox

Acanthamoeba spp.

